

LESSON PLAN TEMPLATE

Lesson Title CAD production of working drawings (006)		Lesson duration 80 min	Stage/year 11 Supervising teacher Mr Wood	Class/Group IPF (Ind. Des. Furniture)	Date Tue 12/5 P5-6
Rationale Practice skills of producing working drawings in CAD. These working drawings will then be basis for cutting lists, work breakdown and time planning.		Syllabus Outcomes P3.1 sketches, produces and interprets drawings in the production of projects P5.2 uses appropriate documentation techniques related to the management of projects		Syllabus Content	
Prior Knowledge Basic computer application knowledge (majority of students have used onShape before, and this should be a refresher). CAD – onShape basic operations (saving, opening, sketching, extruding).			Risk Assessment Class in computer room, ergonomics – seating, hand and eye strain, trips, electrical risks. Remind to setup seats correctly, regular short 1 min break every 15-20 minutes. Bags in middle out of walk areas.		Resources Computers Production folio (word) OnShape (browser) Drawings - cutting list document (to copy) Table in pieces (to examine) Measuring equipment
Specific teaching target – Demonstrate process for creating shape of legs in CAD. <i>(Note: amended from sequence, this class no longer has intro to working drawings, now Lesson 008)</i>					
Learning Intentions	Time	Content/Learning Experience	Teaching Strategies <i>including differentiation</i>	Class Organisation	Assessment Techniques
Explain content of lesson, prepare students for learning (and producing)	10 min	<i>Introduction</i> <i>Write learning intention on board:</i> <i>i. Complete CAD modeling of legs and apron</i> <i>ii. Start modeling tabletop</i> <i>iii. Think about “work breakdown” – how the legs and apron will be produced</i> <i>iv. Write production journal entry for day’s work (last 10 min)</i> Open project portfolio Open onShape and project	 Short discussion, asking what happened last week. How each task in learning intention can be completed.	<i>Initially sitting at desks in middle of class</i> <i>Then move to computers (already turned on)</i>	Remembering what was done last week, Identifying how they feel about the task.
Demonstrate for follow along using Apron piece with tenons.	15 min	• New Part Studio • Sketch – draw width (51), length of piece (475) • Extrude thickness (19) • Transform – select Rotate, Edge of Extrude, Angle (12) • Sketch leg “mortice” (44 x wider than piece) • Extrude – Remove x2 (calculate offset for second, to leave 8mm tennon) • ... this leaves other side to replicate tenon	Think-out-loud Explicit instruction Checking students are following the cutting list document asking about what info it says. And following the reasoning.	Sitting at computer, watching demonstration on front computer Following along on their computers.	Students following, or ahead of instruction. Asking questions about next steps, or issues they are having.
Student conduct production independently	15 min	Have drawing available on board for students to investigate – answer questions. Utilise “features log” (left panel) to step through drawing process.	More capable work Independently, Support students requiring		Asking questions Working on

			assistance with small group guidance		drawing Apron
Demonstrate follow along for Apron halving joint	5 min	Get students to discuss / sketch geometry of having slot on board. How to duplicate a Part Studio (Right click "duplicate") Disucss how to accomplish it - Find centre - Sketch 19mm wide - 12 deg parallelogram (or rotate Apron on 2nd axis and sketch square), - sketch and extrude – Remove 55mm (from top) - On copy use -12 deg parallelogram, offset to other side of apron - Sketch and extrude - Remove 55mm (from bottom)	Group discussion Demonstration for students requiring assistance		Contributing to discussion about how the halving joints are different.
Test angles of halving join "marry up" using Assembly	5 min	Create assembly Add both Apron pieces Use Mate Features Connecting edge – make sure to zoom in to check correct edges selected	Explicit teaching - demonstration	Students watching smart board Possibly following along on own computers.	Students asking clarifying questions, or visibly noting down process to follow
Independent work on CAD production	20 min	Students working on completing Apron, Leg or table top Assemblies	Student conferences - Checking work of students, to ensure they are considering how there are 2 distinct apron pieces and how leg pieces are produced for the 12deg angle etc. Probing questions to guide towards making progress. Peer teaching (if some students ahead to assist others in class)	Smart board for reference (following the features log) Small group discussions Teacher cycling class for conferences	
Write in Production Journal	10 min	Short 2 min discussion asking students what they achieved today, 2 or 3 respond. Then remind of the Production journal prompt "What you did, why you did it, what it means (what will you do next/in the future because of it)" 5 minutes time to write entry 2 min to pack up			
Reflection:					